Design Calculation Sheet - VBF-T6LC-CALC-01

Inputs

Parameter	Value	Source
Positive electrode thickness [m]	5.7e-05	params_final_h45.json
Negative electrode thickness [m]	0.00011	params_final_h45.json
Separator thickness [m]	8e-06	params_final_h45.json
Positive current collector thickness [m]	1e-05	params_final_h45.json
Negative current collector thickness [m]	8e-06	params_final_h45.json
Electrode area [m2]	0.1027	Electrode height x width (0.065 x 1.58)
Positive porosity	0.335	Chen2020
Negative porosity	0.25	Chen2020
Separator porosity	0.47	Chen2020
Positive density [kg/m3]	4400	LNMO.json
Negative density [kg/m3]	1657	Chen2020
Total heat transfer coefficient [W/m2/K]	45	params_final_h45.json
SEI kinetic rate constant [m/s]	1e-15	params_final_h45.json

CapacityEnergy

Metric	Value	Source
1C discharge capacity [Ah]	4.804	final_1c_dfn_h45.json:capacity_ah
Discharge energy [Wh]	19.935	final_1c_energy_dfn_h45.json:energy_wh
Voltage plateau (midpoint) [V]	4.115	final_1c_energy_dfn_h45.json:midpoint_voltage_v
T_max 4C/45C [K]	322.59	final_4c_safety_h45_dfn.json:T_max_K
Anode potential min [V]	0.0894	final_4c_safety_h45_dfn.json:anode_potential_v
SEI thickness 100cyc (DFN) [nm]	9.46	final_aging_dfn_h45.json:sei_thickness_nm_end

EnergyDensity

Metric	Value	Formula / source
Volume [m3]	1.982e-05	total thickness x area
Total thickness [m]	0.000193	sum of 5 layers
Volumetric ED [Wh/L]	1005.77	energy_wh / volume_m3 / 1000
Mass (stack) [kg]	0.041475	calc-energy mass_kg
Gravimetric ED [Wh/kg]	480.66	energy_wh / mass_kg

NPRatio

Quantity	Value	Source
Positive areal capacity density (th x af x Cmax)	1.629915	57um x 0.665 x 43000 mol/m3
Negative areal capacity density (th x af x Cmax)	2.7334725	110um x 0.75 x 33133 mol/m3
N/P ratio	1.677	negative / positive
Stack mass [kg]	0.041475	calc-energy

Process

Parameter	Value	Formula / source
Positive areal density [g/m2]	166.8	57um x (1-0.335) x 4400 kg/m3
Negative areal density [g/m2]	136.7	110um x (1-0.25) x 1657 kg/m3

Parameter	Value	Formula / source
Positive compaction density [g/cm3]	2.926	4400 x (1-0.335)/1000
Negative compaction density [g/cm3]	1.243	1657 x (1-0.25)/1000
Electrolyte fill [g]	6.21	pore volume x 1.2 g/cm3
Formation	0.1C CC to 4.2V, 25C, 2 cycles	design recommended value